mitrafield

Explain it once.
Every beat finds its engine.

An AI director that writes explainer videos and sends every beat to the engine that can show it truthfully: exact math, real maps, charts and diagrams where accuracy matters, Veo and Nano Banana where mood does.

Google Cloud AI Builder Cup 2026 · Theme: **Media, Content & Digital Experiences**

Viraj Sawad · github.com/VirajSawad1021/MitraField

THE PROBLEM

Explainer videos are how the world learns.
Making a correct one is still slow, expensive and manual.

days

of work per minute

A good explainer needs a script, research, animation, maps, charts, narration and editing. Teachers, NGOs and small creators cannot afford a studio.

wrong

numbers drawn by pixels

Text-to-video tools make beautiful clips, but they hallucinate. Coastlines are invented, figures are garbled, equations are squiggles. For learning, a wrong number is worse than none.

one

language, one format

Most explainers exist only in English and only as long-form. Re-making them as Shorts or in Hindi, Nepali or Bahasa means starting again.

THE SOLUTION

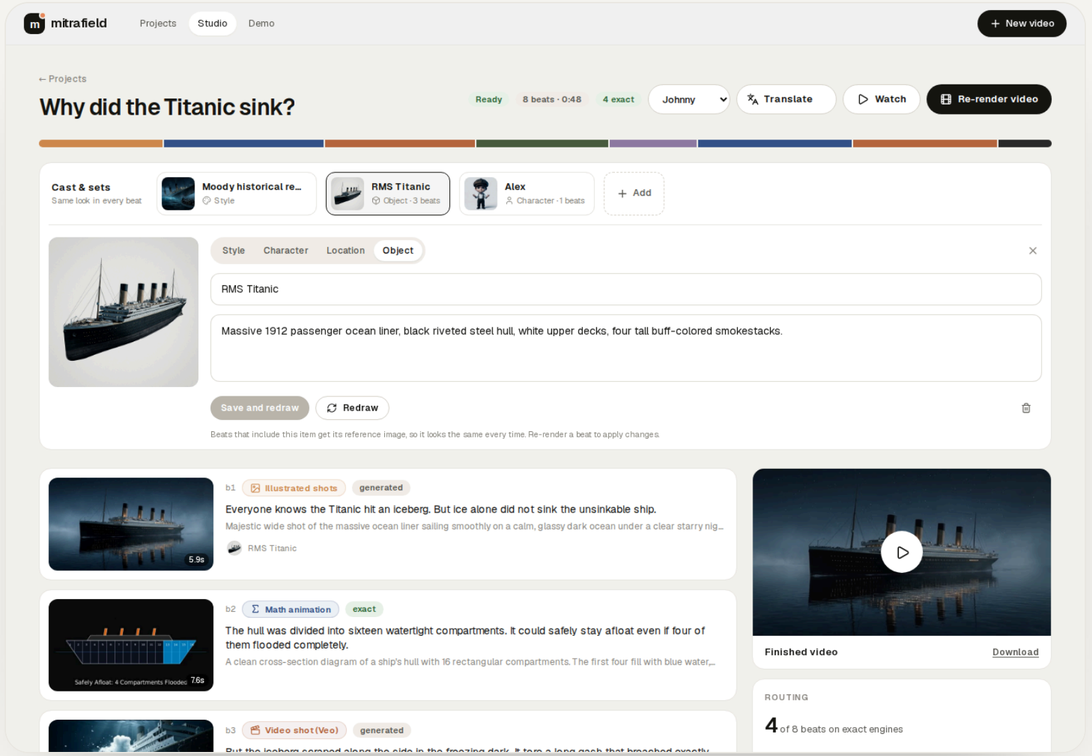
A director agent with eleven engines and ten looks.
Precise where it matters, generative where it helps.

You ask a question and add notes or PDFs. Gemini writes a storyboard for the ear, beat by beat, and routes each beat to the engine that can show it truthfully. Every beat is rendered to the exact length of its narration, checked, and assembled into a finished video with music and subtitles.

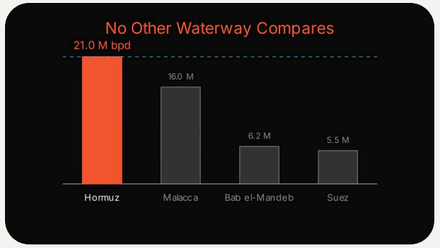
exact Motion scene · Math animation · Globe & maps · Data chart · Timeline · Flow diagram · Whiteboard · Kinetic type

generated Cinematic shots · Presenter · Veo video

Facts, numbers, maps and math are always drawn by code from real data. Ten looks (Cinematic, Editorial flat, Premium 3D, Ink & wash, Motion graphics, Archival, Woodblock, Paper collage, Isometric, Blueprint) restyle every engine at once.



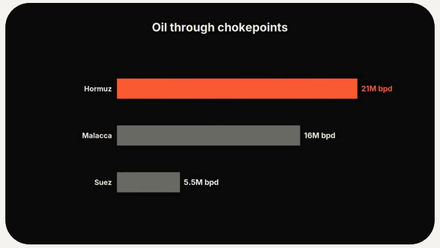
Every engine plans a beat and renders it to the exact length.



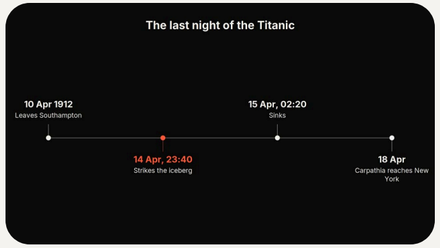
Math animation
Gemini writes Manim code, repairs errors, checks frames



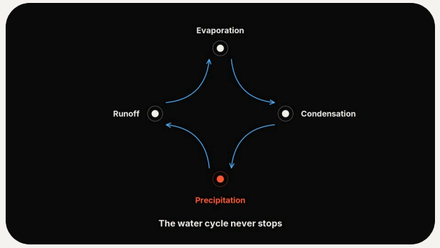
Globe & maps
Natural Earth borders, flights, pins, routes



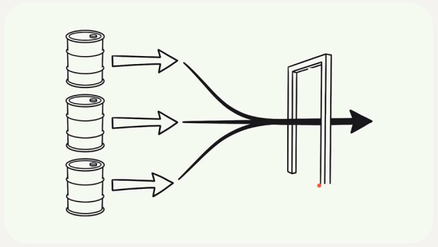
Data chart
Bars and trends from source numbers



Timeline
Events at their real dates



Flow diagram
Processes, cycles, trees



Whiteboard
Line art drawn on stroke by stroke



Kinetic type
Hooks and key numbers



Cinematic shots
Photographic stills in one grade



Presenter
One real-looking person, same face every beat

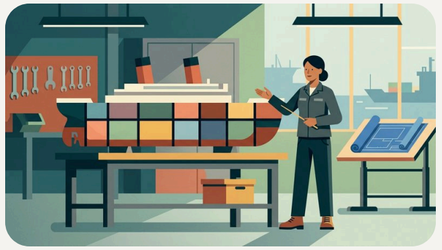


Veo shots
Hero motion, max two per video

One story, ten looks. Professional in every one.



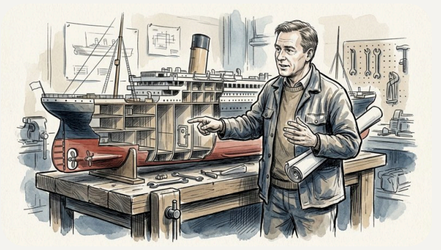
Cinematic



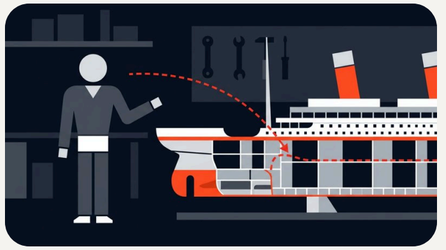
Editorial flat



Premium 3D



Ink & wash



Motion graphics



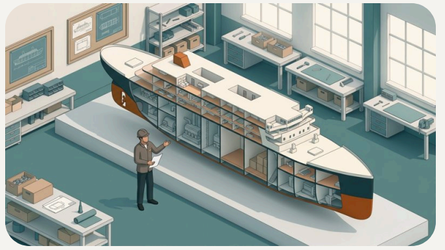
Archival



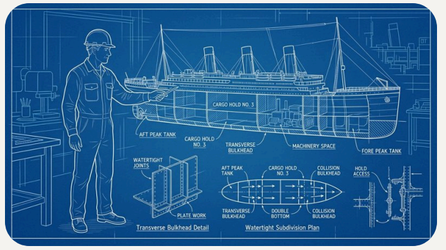
Woodblock



Paper collage



Isometric



Blueprint

A look sets the style of every generated image and its character rules, and gives code-drawn engines its palette and paper texture, so the whole video matches. Same subject shown in each.

HOW IT WORKS

From a question to a finished video.
You only make the calls.

01	**Ask**	YOU	A question plus optional notes, PDFs or images. Format, length, language, narrator, audience.
02	**Storyboard + cast**	GEMINI 3.1 PRO	Narration beat by beat, an engine per beat, and the recurring cast: style, characters, objects, places.
03	**Narrate**	GEMINI TTS	Each line is voiced first, then transcribed and retaken if it drifts from the script.
04	**Render + self-check**	ENGINES	Beats render in parallel to their exact length. Manim code is repaired on error and inspected by Gemini vision.
05	**Edit one beat**	YOU + DIRECTOR	Rewrite a line, swap an engine, or say "show it on a map". Only that beat re-renders.
06	**Assemble + translate**	LYRIA · FFMPEG	Music ducked under the voice, -14 LUFS, subtitles. One-click versions in other languages reuse the visuals.

Define it once. The same ship in every shot.

The director defines the recurring ingredients with the storyboard. Each gets one reference image; every generated beat is drawn from the references it contains, and Veo animates an on-model first frame.



RMS Titanic
object · black hull, white decks, four buff funnels



Thomas
presenter · 1912 Harland and Wolff shipbuilder



Moody historical realism
style · applied to every generated beat



Beat 1 · illustrated still



Beat 3 · Veo video



Beat 7 · Veo video



Beat 5 · presenter

ARCHITECTURE ON GOOGLE CLOUD

Stateless API, durable jobs, a GPU voice that scales to zero.

Firebase Hosting
CDN for the Next.js workspace, security headers, routes /api to Cloud Run

Cloud Run: API
Public, stateless, scales to zero. Passcode, spend caps ($25/day, $5/video), enqueues jobs, never renders

→

Cloud Tasks
Durable queue: retries, backoff, rate limit; jobs survive deploys and restarts

Cloud Run: worker (private)
One job per instance, up to 6 in parallel. Directs, narrates, renders one beat each, assembles, runs QA and fact checks. Model-written code runs in a sandbox

Firestore · Cloud Storage · Secret Manager
Project state with transactions (assembly runs exactly once) · media bucket · API keys

→

Gemini 3.1 Pro + Flash
director, code, vision QA, fact check with Google Search

Nano Banana · Veo 3.1
stills, cast references, hero video

Gemini TTS · Lyria
narration fallback, music bed

Cloud Run GPU (L4)
OmniVoice cloned narrators, scales to zero

Every push to main runs lint, 104 tests (including the Firestore emulator), the web build and the image build, then a gated canary deploy.

TRUST AND SAFETY

Every video checks itself before you see it.

Quality check

After every render: ffmpeg checks length, black or frozen frames, silence and loudness, a contact sheet is saved, and Gemini reviews every beat for cut-off text, garbled words in images and off-model characters. One click fixes a flagged beat.

Fact-check agent

Every number and claim, in the narration and on screen, is checked against the user's sources first, then Google Search, with citations. Contradicted claims are flagged with a correction the director can apply.

No deepfakes

Requests to put words in a real person's mouth are refused. Every image and video prompt is screened: real people become anonymous stand-ins, and video of a real person is never generated.

Prompt-injection guard

Uploaded PDFs and notes are fenced off as data. A test note that said "ignore all previous instructions" and tried to turn the video into a crypto ad was ignored, while its real facts were used.

Spend control

Every model call is priced and logged per video. Hard caps of $25 a day and $5 a video, a passcode on anything that spends, and a cheaper draft mode.

Sandboxed code

Manim code written by the model runs in a sandbox without network access, and motion scenes render offline in headless Chrome.

GEN AI IMPLEMENTATION

Not one prompt. Seven loops that check each other.

01

Route

The director reads every engine's description and picks per beat, under rules: facts go to exact engines, Veo is capped at two.

02

Code + repair

Gemini writes Manim for the exact duration. Render errors go back with the traceback until it compiles.

03

Look + fix

Gemini vision inspects rendered frames for overlapping or cut-off text and asks for a corrected scene.

04

Listen + retake

Every narration line is transcribed and retaken if words were added, dropped or garbled.

05

Stay on model

Cast references feed every image and Veo first frame, so characters and subjects stay consistent.

06

Review the cut

After assembly Gemini reviews every beat of the finished video and ffmpeg checks picture and sound; a flagged beat is fixed in one click.

07

Check the facts

Claims are verified against the sources and Google Search; contradicted ones come back to the director with the evidence.

Failures degrade gracefully: a refused Veo clip becomes a still, a failed engine becomes a title card, and the video still renders.

RESULTS FROM THE PROTOTYPE

Finished explainers from a single prompt.

10

finished videos: geography, math, biology, history, tech, anatomy and earth science, in 3 looks and 3 languages

2–9 min

to render a 45–60 s video, unattended, no manual edits

$0.72–1.70

measured API cost per video, logged per call; a translation that reuses the shots cost $0.34

56%

of all beats rendered by exact engines, so facts are computed, not generated



Woodblock look · tsunami



Editorial flat · vaccine Short with a cast presenter



Hindi translation · labels re-rendered in Devanagari

MitraField · costs from Google's published Gemini API rates, logged per call

IMPACT

Correct explainers for everyone who teaches.
In their language, in their format.

Teachers and schools

Turn a lesson plan into a narrated video with accurate diagrams, then into a Short for revision. One teacher, minutes, a few dollars.

NGOs and newsrooms

Explain a policy, a disease or a crisis with real maps and real numbers, cited on screen, and publish it in several JAPAC languages the same day.

Creators

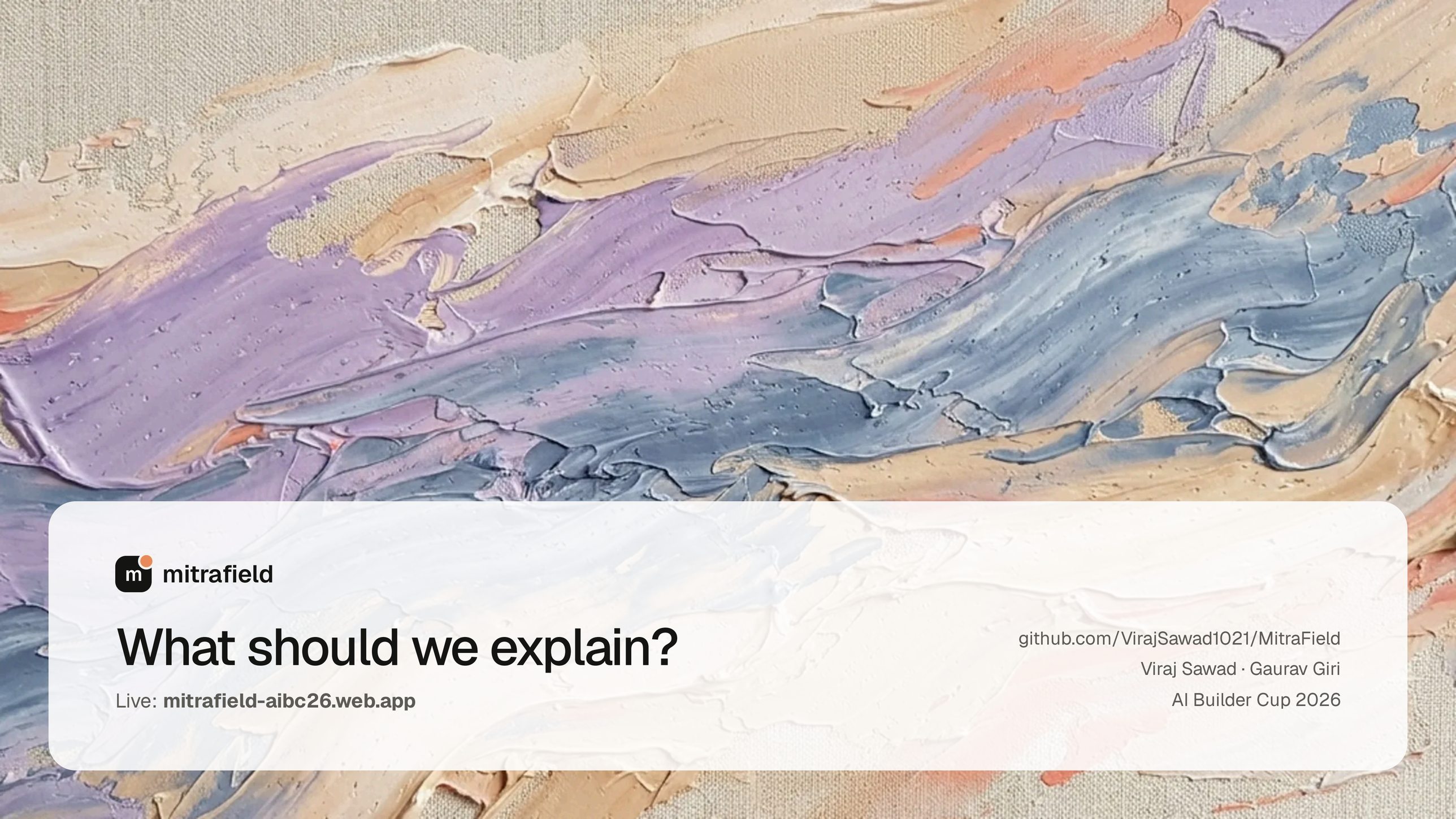
Run an explainer channel with a consistent presenter and style across every episode, long-form and Shorts from one workspace.

Misinformation angle: every number and map in a MitraField video is computed from data the user can see, not invented by a pixel model.

FEASIBILITY AND SCALE

Built to run cheaply, and to grow.

AREA	TODAY	NEXT
Compute	API and worker on Cloud Run, Cloud Tasks queue, one beat per worker instance, Firestore state, content-hash cache	Autoscaled worker pools per tenant, render priority lanes
Cost control	Per-call ledger, $25/day and $5/video caps, Veo length fitted to the beat, draft mode on Veo Lite	Batch API for final renders, per-user budgets
Styles	Ten engines behind one contract, discovered automatically	Scale comparisons, source-quote cards, code walkthroughs, real archival footage
Learning	Audience and level steer the script	Learn mode: quizzes after the video, beats re-explained where viewers struggle, analytics in BigQuery
Languages	One-click translation that reuses the visuals	Cloud Translation glossaries, regional voices



mitrafield

What should we explain?

Live: **mitrafield-aibc26.web.app**

github.com/VirajSawad1021/MitraField

Viraj Sawad · Gaurav Giri

AI Builder Cup 2026